

Syeduzzaman Khan

Danbury, CT

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience designing, developing, and deploying scalable, high-performance applications across cloud-native and data-driven environments. Proven expertise in backend engineering (Python, FastAPI, Django), large-scale ETL pipelines, distributed systems, and full-stack development using React and TypeScript. Strong background in AWS architecture, database optimization, and observability with OpenTelemetry and New Relic. Experienced in building secure, resilient systems serving high-volume financial and enterprise workloads.

TECHNICAL SKILLS

- Languages: Python, TypeScript, JavaScript, SQL
- Backend: FastAPI, Django, Flask, REST, GraphQL, Pydantic
- Frontend: React, TypeScript, MobX, Redux, Angular, HTML, CSS
- Databases: PostgreSQL, MySQL, Amazon RDS, DynamoDB, Snowflake
- Data & ETL: AWS Glue (Spark), pandas, NumPy, EMR, Athena, Redshift
- Cloud & DevOps: AWS (Lambda, S3, EC2, RDS, Fargate, IAM, CloudFormation, SNS, SQS), Docker, Kubernetes
- Monitoring: OpenTelemetry, New Relic, Grafana, CloudWatch
- CI/CD: GitHub, GitLab, Terraform, CodePipeline, CodeBuild
- Other: Redis, Unix/Linux, JSON/XML processing, Agile/Scrum

PROFESSIONAL EXPERIENCE

Capital One via TekSystems — New York, NY, Full Stack Developer | August 2024 – Present

- Contribute to the Enterprise Data Detection platform enabling automated discovery and classification of sensitive PII data across enterprise systems.
- Designed and owned scalable AWS Glue (Spark) ETL pipelines processing GB-scale datasets from S3 and Snowflake/OneLake into Amazon RDS with idempotent upsert logic.
- Built serverless FastAPI microservices deployed on AWS Lambda delivering highly available and cost-efficient APIs.
- Developed high-performance REST APIs serving millions of records with optimized pagination, filtering, and query strategies.
- Implemented strong data validation using Pydantic models ensuring schema integrity and reliable downstream processing.
- Enhanced frontend experience (Metabot Disposition Experience) using React, TypeScript, and MobX.
- Built optimized React data tables handling millions of rows with advanced filtering, search, multi-select, and expansion features.

- Improved PostgreSQL performance via indexing, partitioning, materialized views, and query optimization strategies.
- Implemented distributed tracing and metrics using OpenTelemetry and New Relic to improve system reliability and observability.
- Delivered secure cloud-native architecture leveraging RDS, S3, Lambda, Route53, and Fargate.

Goldman Sachs via ConsultAdd— New York, NY, Backend Data Engineer | June 2024 – August 2024

- Designed and implemented an Equity Margin Loan data pipeline supporting mission-critical financial workflows.
- Developed APIs using Python and internal SLANG language for financial business logic processing.
- Architected scalable ETL pipelines loading structured and unstructured data into Sybase IQ and Snowflake.
- Performed deep data validation and anomaly detection using pandas and NumPy.
- Deployed production services via Terraform and enterprise monitoring frameworks.
- Collaborated with risk, compliance, and engineering teams to deliver production-grade data solutions.

Nutrien Ag Solutions via Ioneq Inc— Loveland, CO, Senior Software Engineer | June 2021 – March 2024

- Designed and developed scalable Agri-tech platforms supporting crop planning and agricultural analytics.
- Built secure backend services using Django and Python powering web and mobile applications.
- Delivered full-stack features using React, TypeScript, Angular, Redux, GraphQL, and Material UI.
- Optimized PostgreSQL and DynamoDB data models for high-volume datasets.
- Deployed cloud-native solutions using AWS EC2, S3, RDS, Lambda, Elasticsearch, and API Gateway.
- Implemented CI/CD pipelines with CodePipeline and CodeBuild.
- Developed containerized microservices using Docker and Kubernetes.
- Applied security best practices including authentication, authorization, and secure data handling.

United Commercial Bank — Bangladesh, Software Developer | August 2014 – July 2018

- Developed REST APIs using Flask and SQLAlchemy for data center management systems.
- Built frontend applications using React, JavaScript, and HTML/CSS.
- Optimized backend performance using Python functional constructs and efficient bulk data operations.

EDUCATION

Master of Science in Engineering Science (Computer Engineering)
University of the Pacific, California